



ichosynth — Development Environment

Complete, up-to-date guide for Windows and macOS

Everything you need to build, modify, and upload the firmware: from the toolchain to the libraries, all the way to your first upload onto the Teensy 4.1.

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arduino-cli required

teensy:avr ≥ 1.61.0

OS Windows · macOS

⚡ **In a hurry?** ichosynth runs the real **TœRN** firmware, and one command builds it for our hardware: `python teensy/build_toern.py`. Jump to [chapter 3](#). The rest of the guide explains every step and prerequisite.



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1 · What we install

ichosynth is a **Teensy 4.1** running the real **TœRN** firmware (by SP_ / soundpauli — <https://toern.live>) on cheap, hand-soldered parts. We don't keep a separate copy of TœRN in the repo: a build script fetches the upstream sources and compiles them for our hardware. Here's what you need:

Component	Version	What it's for	Required
arduino-cli	latest	the command-line build toolchain	✓
teensy:avr core	≥ 1.61.0	the Teensy compiler + Teensy Loader	✓
Python	3.x	runs <code>teensy/build_toern.py</code> (the build script)	✓

Component	Version	What it's for	Required
Git	latest	the script clones the TCERN sources on first run	✓
Custom drivers	in the repo	i2cEncoderLibV2 , FastTouch , Ichos01ed — already in teensy/libraries/	✓ (bundled)
GitHub CLI	latest	optional, for version-controlling the project	optional

⚙️ **The one thing that bites everyone:** TCERN is a single ~23k-line translation unit, and the Teensy gcc **crashes at the default -O2** (exits 0xffffffff — an internal compiler error / out-of-memory). The fix is to compile at **-O1** (`opt=o1std`). The build script sets this for you; if you build from the Arduino IDE instead, you must pick **Tools** → **Optimize** → **"Fast"**. See [chapter 7](#).

2 · Requirements

- **Windows:** Windows 10 or 11, 64-bit.
- **macOS:** macOS 10.15 or later; native support for **Intel** and **Apple Silicon** (M1/M2/M3/M4).
- **A micro-USB data cable** (not charge-only) for the Teensy 4.1.
- A **Teensy 4.1 with 16 MB PSRAM** (both chips) soldered on — TCERN needs the PSRAM for its sample buffers.
- ~1 GB of disk space for the toolchain + core + libraries.

3 · The fast track: one command builds it all

Once you have **arduino-cli**, the **teensy:avr core**, **Python**, and **Git** (chapters 4–5), building the firmware is a single command from the repo root:

```
python teensy/build_toern.py
```

This produces the firmware at `teensy/firmware/toern.hex` . To build *and* flash in one go (if `teensy_loader_cli` is on your PATH):

```
python teensy/build_toern.py --flash
```

Useful flags:

Flag	Effect
<code>--flash</code>	after building, upload with <code>teensy_loader_cli</code>
<code>--keep</code>	keep the temporary build folder (for inspecting the staged/patched sources)

💡 The **first** build is slow (the Teensy core is compiled once) and it also **clones the TCERN sources** into `emulator/toern-src/` if they aren't there yet. Later builds are fast.

🔔 Just want to flash a **prebuilt** `.hex` ? There's a one-click GUI flasher in `_FLASHER/` — see [chapter 8.3](#).

4 · Installing the toolchain (arduino-cli)

We build from the command line with `arduino-cli`.

Windows

```
winget install ArduinoSA.CLI
```

macOS

```
brew install arduino-cli
```

Check it's on your PATH:

```
arduino-cli version
```

i Drivers: on Windows 10/11 and on macOS **no additional drivers are needed** for the Teensy 4.1.

5 · Adding Teensy support (the teensy:avr core)

The Teensy compiler and Teensy Loader come from the **teensy:avr core**. Add PJRC's package index, update, and install the core (we need **version ≥ 1.61.0**):

```
arduino-cli config init
arduino-cli config add board_manager.additional_urls https://www.pjrc.com/teensy/package_teensy_index.json
arduino-cli core update-index
arduino-cli core install teensy:avr
```

Check the version you got:

```
arduino-cli core list
```

You should see `teensy:avr` at **1.61.0** or newer.

💡 Prefer a GUI? You can instead install **Teensyduino 1.61** through the Arduino IDE 2.x Boards Manager (PJRC's package URL above) — that installs the same `teensy:avr` core. Either way, the build script just needs the core present

and `arduino-cli` on your PATH.

6 · The libraries

The TCERN firmware compiles against three **custom drivers** that adapt it to our cheap hardware. They are **already in the repo** under `teensy/libraries/`, and the build script points `arduino-cli` at them automatically — you don't install them:

Library	Replaces	Where the hardware config lives
i2cEncoderLibV2	TCERN's 4 I ² C RGB encoders → our 4 KY-040 mechanical encoders	ICHOS_ENC_PINS in <code>i2cEncoderLibV2.h</code>
FastTouch	TCERN's 3 capacitive touch pads → our 3 momentary tact switches	ICHOS_BTN_PINS in <code>FastTouch.h</code>
IchosOled	adds the I ² C OLED HUD (the state TCERN showed on the encoder RGB rings)	<code>teensy/libraries/IchosOled/</code>

 **Where the hardware lives:** pin assignments live in those two headers (`ICHOS_ENC_PINS` , `ICHOS_BTN_PINS`) and in `teensy/build_toern.py` — **not** in a `config.h` . (`config.h` belongs to the old NI404 fallback build and is not used by the TCERN port.)

The build script also resolves any remaining stock libraries (the **Audio**, **Wire**, **SD**, etc. bundled in the Teensy core, plus anything in your user `Arduino/libraries/`) the way `arduino-cli` normally does.

7 · How the build script works

`teensy/build_toern.py` compiles the **unmodified** TCERN sketch, applying only the minimal changes that make it run on our hardware. Step by step:

1. **Locate the sources.** It looks for the TCERN sketch under `emulator/toern-src/` ; if it isn't there, it **clones it** from upstream automatically (so the repo stays free of a vendored copy).
2. **Stage them** into a temporary build folder (the originals stay pristine).
3. **Pin remap.** TCERN's `SWITCH_1/2/3` default to pins that collide with our right-hand KY-040, so they are remapped to **25 / 26 / 28** — the same pins as `ICHOS_BTN_PINS` in `FastTouch.h` .
4. **Feature trim.** It removes the optional **2nd LED strip** (frees a pin and saves CPU), matching our build.
5. **OLED HUD.** It wires in the `IchosOled` library (include + `begin()` + a one-line render call in `loop()`) so the status HUD compiles as its own translation unit.
6. **Compile** with `arduino-cli` for the FQBN below and copy the result to a stable name, `toern.hex` .

The exact target it builds for is:

```
teensy:avr:teensy41:usb=serialmidi16,opt=o1std
```

- `usb=serialmidi16` → **USB type Serial + MIDIx16** (so `usbMIDI` works).

- `opt=01std` → `-01` . This is the critical flag: the default `-02` crashes the Teensy gcc on TCERN's giant single translation unit. `-01` builds cleanly and the firmware still fits with room to spare.

8 · Building and uploading

8.1 Build

From the repo root:

```
python teensy/build_toern.py
```

When it finishes you'll have `teensy/firmware/toern.hex` .

8.2 Build and flash in one step

If you have `teensy_loader_cli` on your PATH:

```
python teensy/build_toern.py --flash
```

1. Connect the Teensy 4.1 via USB (a **data** cable).
2. The loader usually programs on its own; if it stays waiting, press the **little white button** on the Teensy.
3. When the flash finishes, the animation starts on the LED matrix.

8.3 Flashing the firmware

If you'd rather flash a **prebuilt** `.hex` (or don't have `teensy_loader_cli`), use the **one-click GUI flasher** in `_FLASHER/` — point it at `teensy/firmware/toern.hex` and click flash.

9 · Building from the Arduino IDE (alternative)

You can also build TCERN by hand in the Arduino IDE 2.x — useful for stepping through the code. You'll need to reproduce what the script does:

Tools menu item	Value
Board	Teensy 4.1
USB Type	Serial + MIDIx16
CPU Speed	600 MHz (default)
Optimize	"Fast" ⚠️ (this is <code>-01</code> ; the default "Faster"/ <code>-02</code> crashes the compiler)
Port	the Teensy's port (after connecting it)

You'll also have to apply the pin remap (SWITCH_1/2/3 → 25/26/28), make the three custom libraries in teensy/libraries/ visible to the IDE, and wire in the OLED HUD by hand. **The script does all of this for you**, which is why it's the recommended path.

 Don't forget **Optimize** → "**Fast**". If you leave it at the default, the build dies with an internal compiler error (`exit 0xffffffff`) partway through TCERN's single huge file.

10 · Optional tools (Python, Git)

10.1 Python

The build script needs **Python 3**. (Git is also required — see below — because the script clones the TCERN sources on first run.)

Windows: `winget install Python.Python.3.12` · **macOS:** `brew install python@3.12`

 Any modern Python 3 works for the build script. (If you also use the wavmaker sample converter and run it from source on **Python 3.13+**, that one needs `pip install audioop-lts`, because `audioop` was removed from the standard library — but the ready-made `wavmaker.exe` in `_SDCARD/` needs no Python at all.)

10.2 Git and GitHub CLI

Git is required (the script clones TCERN). GitHub CLI is optional.

Windows: `winget install Git.Git` (+ `GitHub.cli` optional) **macOS:** `xcode-select --install` (includes git) · `brew install gh` (optional)

11 · Checklist and common problems

✓	Check
☐	<code>arduino-cli version</code> works (it's on your PATH)
☐	<code>arduino-cli core list</code> shows teensy:avr ≥ 1.61.0
☐	Python 3 and Git are installed
☐	<code>python teensy/build_toern.py</code> produces <code>teensy/firmware/toern.hex</code>
☐	(IDE build only) USB Type = Serial + MIDIx16 and Optimize = "Fast"
☐	Teensy 4.1 has 16 MB PSRAM soldered

Symptom	Cause / fix
<code>arduino-cli</code> not on PATH ... from the script	install <code>arduino-cli</code> (ch. 4) and reopen the terminal
Compile dies with <code>exit 0xffffffff</code> / internal compiler error	you built at <code>-02</code> ; use the script, or set IDE Optimize → "Fast" (<code>-01</code>)
<code>teensy:avr</code> missing / too old	run <code>arduino-cli core install teensy:avr</code> and check <code>core list</code> (need $\geq 1.61.0$)
Script can't clone the TERN sources	install Git, or clone manually into <code>emulator/toern-src/</code>
'usbMIDI' was not declared (IDE build)	USB Type not set to Serial + MIDIx16
<code>teensy_loader_cli</code> not found ON <code>--flash</code>	use the GUI flasher in <code>_FLASHER/</code> instead
Teensy not programming	press the white button on the board; or use a data USB cable, not charge-only
Port not visible	charge-only USB cable → use a data cable

Part of the *ichosynth* project · runs the real **TERN** (SP_) firmware · for the ICHOS 2026 workshop.